The Æther-flow Interpretation of Relativity

Substrate-Response / Self-Response Charge-Polarization Candidate
Exact-Preservation Root Shell / Lifted Connection-Cotangent / Parallel-Transport
Origin Note

Abstract

The exact-preservation pre-ambient-shell / connection-cotangent / parallel-transport origin note already shows that the previously recorded pre-ambient layer is not primitive at present scope. It is recovered there from deeper root substrate bundles carrying pre-ambient-generating shell sections, lifted ambient-generating shell maps, lifted shell maps and lifted exact-constraint maps into the same target metrics, root cotangent-model Liouville one-forms and closed two-form geometry, root connection one-forms and induced parallel transports, root-basic shadow / rate / density / flux / boundary data, and fixed-endpoint root rate-action geometry. What remained open is one layer deeper again: why that root layer itself should exist rather than becoming the present deepest disciplined postulate.

The present note addresses exactly that narrower burden. It keeps fixed the same exact-preservation target: the same $\mathscr{F}_{\mathrm{sel}}$, the same exact-preservation normal form $\mathscr{F}_{\mathrm{deep}}$, the same carrier-origin / primitive-reservoir / microscopic-equilibrium functionals

$$\mathscr{F}_{\mathrm{car}}[\mathfrak{S}, \mathfrak{R}, \mathfrak{F}], \qquad \mathscr{F}_{\mathrm{prim}}[K, J, F], \qquad \mathscr{F}_{\mathrm{mic}}[H, Y],$$

the same Σ^{cmp} solve, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same one operative metric, and the same recorded Phase D / E and Phase F gains. Its positive move is to place the recorded root-shell / lifted connection-cotangent / parallel-transport layer under a still deeper primitive root-generating construction:

1. finite-dimensional exact-preserving primitive substrate bundles with visible readouts, unit sections, time reversals, preserved positive metrics, and compact visible fibers;
2. root-generating shell sections whose zero loci are exactly the recorded root substrate bundles;
3. lifted pre-ambient-generating shell maps, lifted ambient-generating shell maps, lifted shell maps, and lifted exact-constraint maps into the same positive target metrics whose root restrictions are exactly the recorded root shell sections and root lifted maps;
4. cotangent-model Liouville one-forms and closed two-forms on the primitive bundles whose root restrictions are exactly the recorded root Liouville and closed two-form geometry;
5. exact-preserving unitary primitive connection one-forms whose parallel transports along primitive curves restrict exactly to the recorded root connection one-forms and their induced root parallel transports;
6. primitive-basic shadow / rate / density / flux / boundary data and fixed-endpoint primitive rate actions whose root restrictions are exactly the recorded root-basic data and the same fixed-endpoint root rate-action geometry.

At current exact-preservation scope, the root substrate bundles, their pre-ambient-generating shell sections, their lifted ambient-generating shell maps, their lifted shell maps and lifted exact-constraint maps with the same target metrics, their root cotangent-model Liouville one-forms and closed two-form geometry, their root connection one-forms and induced parallel transports, their root-basic shadow / rate / density / flux / boundary data, and their fixed-endpoint root rate-action geometry are therefore no longer primitive either. Because the present note reproduces the full input layer of the immediately preceding pre-ambient-shell / connection-cotangent / parallel-transport origin note without change, that note applies verbatim. Consequently the same pre-ambient-shell / connection-cotangent / parallel-transport layer, the same ambient-shell / cotangent-Liouville / connection-holonomy layer, the same process-state layer, the same reversible process-thread / process-window / history / groupoid / reconfiguration / derivation chain, the same $\mathscr{F}_{\mathrm{sel}}$, the same exact-preservation normal form, the same carrier-origin / primitive-reservoir / microscopic-equilibrium functionals, the same Σ^{cmp} solve, the same $\Pi_{\mathrm{free}}H = 0$ outcome, the same observer-equation closure, the same one operative metric, and the same recorded Phase D / E and Phase F gains are preserved without change.

The scope remains disciplined. This note does not claim a final microscopic completion, a uniqueness theorem for every conceivable deeper realization, or a completed first-principles derivation of GR. Its narrower exact positive claim is only that the recorded root-shell / lifted connection-cotangent / parallel-transport layer is itself no longer primitive at present scope, but is forced by a still deeper primitive root-generating shell / lifted connection-cotangent layer while preserving the same downstream package.

1. Introduction

The active one-metric charge-polarization line has now crossed the candidate, benchmark-gate, controlled-limit, residual-sector, redesign, substrate-anchoring, substrate-action, kernel-origin, deeper-selection, primitive-reservoir, carrier-origin, precursor-selection, exact-preservation normal-form, microphysical-Hessian, charge-generator, derivation-algebra, reconfiguration-group, reversible-groupoid, reversible-history, reversible process-window, reversible process-thread, process-state, proto-process, constraint-shell / Liouville-holonomy, ambient-shell / cotangent-connection, and pre-ambient-shell / connection-cotangent / parallel-transport origin steps on the same GR-directed route. The burden therefore sharpens again. It is not another packaging pass. It is not stronger promotion language. It is not, by default, a move onto the anchored positive-pair side line. It is to go one layer below the currently recorded root-shell / lifted connection-cotangent / parallel-transport construction itself and explain why that construction is forced.

That burden has eight parts. First, one must explain why the exact-preserving root substrate bundles

$$\mathcal{R}_{\mathrm{car}}, \qquad \mathcal{R}_{\mathrm{cur}}, \qquad \mathcal{R}_{\mathrm{hom}}$$

should exist rather than being declared. Second, one must explain why the visible root readouts, root unit sections, root time reversals, preserved positive root metrics, and compact visible root fibers should be forced rather than inserted independently. Third, one must explain why the pre-ambient-generating shell sections

$$\mathfrak{Z}^{\mathrm{root}}_{\mathrm{car}}, \qquad \mathfrak{Z}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \mathfrak{Z}^{\mathrm{root}}_{\mathrm{hom}}$$

should be forced rather than postulated. Fourth, one must explain why the lifted ambient-generating

shell maps

$$\widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{car}}, \qquad \widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{cur}}, \qquad \widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{hom}}$$

should themselves be induced rather than separately chosen. Fifth, one must explain why the lifted shell maps

$$\widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{car}}, \qquad \widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{hom}}$$

and lifted exact-constraint maps

$$\widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{car}}, \qquad \widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{hom}}$$

into the same target metrics should themselves be induced rather than separately chosen. Sixth, one must explain why the root cotangent-model Liouville one-forms

$$\boldsymbol{\Lambda}^{\mathrm{root}}_{\mathrm{car}}, \qquad \boldsymbol{\Lambda}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \boldsymbol{\Lambda}^{\mathrm{root}}_{\mathrm{hom}}$$

their closed two-forms

$$\boldsymbol{\Xi}^{\mathrm{root}}_{\mathrm{car}}, \qquad \boldsymbol{\Xi}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \boldsymbol{\Xi}^{\mathrm{root}}_{\mathrm{hom}}$$

and the root connection one-forms

$$\mathcal{B}^{\mathrm{conn}}_{\mathrm{car}}, \qquad \mathcal{B}^{\mathrm{conn}}_{\mathrm{cur}}, \qquad \mathcal{B}^{\mathrm{conn}}_{\mathrm{hom}}$$

together with their induced parallel transports

$$\mathbb{P}_{\mathrm{car}}, \qquad \mathbb{P}_{\mathrm{cur}}, \qquad \mathbb{P}_{\mathrm{hom}}$$

should themselves be induced rather than separately prescribed. Seventh, one must explain why the root-basic shadow / rate / density / flux / boundary data and the fixed-endpoint root rate-action geometry should be forced rather than added as independent decorations. Eighth, one must re-show that solving these deeper burdens changes none of the already recorded gains: the same pre-ambient-shell / connection-cotangent / parallel-transport layer, the same ambient-shell / cotangent-Liouville / connection-holonomy layer, the same process-state layer, the same reversible process-thread / process-window / history / groupoid / reconfiguration / derivation chain, the same $\mathscr{F}_{\mathrm{sel}}$, the same exact-preservation normal form, the same carrier-origin / primitive-reservoir / microscopic-equilibrium functionals, the same Σ^{cmp} solve, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same observer-equation closure, the same one operative metric, and the same recorded Phase D / E and Phase F gains must remain intact.

The positive claim of the present note is that there is a disciplined way to do exactly that. One introduces a still deeper primitive exact-preserving layer whose visible fibers are compact, whose root-generating shell sections cut out the current root bundles as exact zero shells, whose lifted pre-ambient-generating shell maps, lifted ambient-generating shell maps, lifted shell maps, and lifted exact-constraint maps restrict to the current root maps, whose cotangent-model one-forms descend by shell restriction to the current root Liouville geometry, whose skew-Hermitian primitive connection one-forms restrict to the current root connection one-forms and generate the current root transports by parallel transport, and whose primitive-basic one-forms / source fields / boundary potentials together with the same fixed-endpoint variational structure restrict to the current root-basic data. The immediately preceding pre-ambient-shell / connection-cotangent / parallel-transport note can

then be invoked without alteration, so the whole downstream chain is preserved verbatim.

2. Fixed Inputs from the Recorded Completion Line

Definition 1 (Fixed branch and downstream completion target)**.** *The present note keeps fixed the same charge-polarization branch*

$$g^{\mathrm{cand}} = g^{\mathrm{br}} + \Sigma^{\mathrm{IER}} + \Sigma^{\mathrm{resp}}, \qquad g^{\sharp} = g^{\mathrm{br}} + \Sigma^{\mathrm{IER}} + \Sigma^{\mathrm{resp}} + \Sigma^{\mathrm{cmp}}, \tag{1}$$

with lifted variables satisfying

$$\Xi^{*}\widehat{\Sigma}^{\mathrm{IER}} = \Sigma^{\mathrm{IER}}, \qquad \Xi^{*}\widehat{\Sigma}^{\mathrm{resp}} = \Sigma^{\mathrm{resp}}, \qquad \Xi^{*}\widehat{\Sigma}^{\mathrm{cmp}} = \Sigma^{\mathrm{cmp}}. \tag{2}$$

The same completion class and the same free-sector condition are fixed:

$$\widehat{\Sigma}^{\mathrm{cmp}} \in \mathscr{C}_{\mathrm{cmp}}, \qquad \Pi_{\mathrm{free}}\widehat{\Sigma}^{\mathrm{cmp}} = 0. \tag{3}$$

The same substrate and observer completion equations are fixed as well:

$$\widehat{\mathcal{L}}(\widehat{\Sigma}^{\mathrm{cmp}}) = -\widehat{\mathcal{V}}^{\mathrm{res}}, \qquad \mathcal{L}_{g^{\mathrm{br}}}(\Sigma^{\mathrm{cmp}}) = -\mathcal{V}^{\mathrm{res}}_{\mu\nu}. \tag{4}$$

Every statement below must preserve these equations together with the same one operative metric $g^{\sharp}$.

Definition 2 (Recorded root-shell / lifted connection-cotangent / parallel-transport input layer)**.** *The immediately preceding pre-ambient-shell / connection-cotangent / parallel-transport origin note fixes the following exact-preserving input layer.*

1. *Finite-dimensional exact-preserving root substrate bundles*

$$\mathcal{R}_{\mathrm{car}}, \qquad \mathcal{R}_{\mathrm{cur}}, \qquad \mathcal{R}_{\mathrm{hom}}$$

   with visible root readout maps

$$\Pi^{\mathrm{root}}_{\mathrm{car}}, \qquad \Pi^{\mathrm{root}}_{\mathrm{cur}}, \qquad \Pi^{\mathrm{root}}_{\mathrm{hom}},$$

   exact-preserving root unit sections

$$u^{\mathrm{root}}_{\mathrm{car}}, \qquad u^{\mathrm{root}}_{\mathrm{cur}}, \qquad u^{\mathrm{root}}_{\mathrm{hom}},$$

   involutive root time reversals

$$\varrho^{\mathrm{root}}_{\mathrm{car}}, \qquad \varrho^{\mathrm{root}}_{\mathrm{cur}}, \qquad \varrho^{\mathrm{root}}_{\mathrm{hom}},$$

   and preserved positive root metrics

$$\overline{M}_{\mathrm{car}}, \qquad \overline{M}_{\mathrm{cur}}, \qquad \overline{M}_{\mathrm{hom}}.$$

   Each visible root fiber is compact.

2. *Finite-dimensional real pre-ambient-generating target spaces*

$$\mathcal{Z}^{\mathrm{root}}_{\mathrm{car}}, \qquad \mathcal{Z}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \mathcal{Z}^{\mathrm{root}}_{\mathrm{hom}}$$

with positive metrics

$$p^{\mathrm{root}}_{\mathrm{car}}, \qquad p^{\mathrm{root}}_{\mathrm{cur}}, \qquad p^{\mathrm{root}}_{\mathrm{hom}},$$

and exact-preserving pre-ambient-generating shell sections

$$\mathfrak{Z}^{\mathrm{root}}_{\mathrm{car}}, \qquad \mathfrak{Z}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \mathfrak{Z}^{\mathrm{root}}_{\mathrm{hom}}$$

whose zero loci are exactly the current pre-ambient substrate bundles. These sections carry the same recorded positive root gap bounds $\kappa^{\mathrm{root}}_{\mathrm{car}}, \kappa^{\mathrm{root}}_{\mathrm{cur}}, \kappa^{\mathrm{root}}_{\mathrm{hom}} > 0$ *on those current pre-ambient shells.*

3. *Exact-preserving lifted ambient-generating shell maps*

$$\widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{car}}, \qquad \widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{cur}}, \qquad \widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{hom}}$$

into the same positive target spaces $\mathcal{H}^{\mathrm{orig}}_{\mathrm{car}}, \mathcal{H}^{\mathrm{orig}}_{\mathrm{cur}}, \mathcal{H}^{\mathrm{orig}}_{\mathrm{hom}}$ *with the same positive target metrics* $n^{\mathrm{orig}}_{\mathrm{car}}, n^{\mathrm{orig}}_{\mathrm{cur}}, n^{\mathrm{orig}}_{\mathrm{hom}}$*. Their restrictions to the current pre-ambient shells are exactly the recorded ambient-generating shell sections and carry the same recorded positive ambient-generating gap bounds* $\epsilon^{\mathrm{orig}}_{\mathrm{car}}, \epsilon^{\mathrm{orig}}_{\mathrm{cur}}, \epsilon^{\mathrm{orig}}_{\mathrm{hom}} > 0$ *on the current ambient shells.*

4. *Exact-preserving lifted shell maps*

$$\widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{car}}, \qquad \widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{hom}}$$

into the same positive shell target spaces $\mathcal{E}_{\mathrm{car}}, \mathcal{E}_{\mathrm{cur}}, \mathcal{E}_{\mathrm{hom}}$ *with the same positive shell target metrics* $m^{\mathrm{sh}}_{\mathrm{car}}, m^{\mathrm{sh}}_{\mathrm{cur}}, m^{\mathrm{sh}}_{\mathrm{hom}}$*. Their restrictions to the current pre-ambient exact-constraint shells are exactly the recorded lifted shell maps and carry the same recorded positive shell-gap bounds* $\eta_{\mathrm{car}}, \eta_{\mathrm{cur}}, \eta_{\mathrm{hom}} > 0$ *on the current seed shells.*

5. *Exact-preserving lifted exact-constraint maps*

$$\widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{car}}, \qquad \widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{hom}}$$

into the same positive target spaces $\mathcal{K}_{\mathrm{car}}, \mathcal{K}_{\mathrm{cur}}, \mathcal{K}_{\mathrm{hom}}$ *with the same positive target metrics* $m^{\mathrm{con}}_{\mathrm{car}}, m^{\mathrm{con}}_{\mathrm{cur}}, m^{\mathrm{con}}_{\mathrm{hom}}$*. Their restrictions to the current pre-ambient shells are exactly the recorded lifted exact-constraint maps and carry the same recorded positive exact-constraint gap bounds* $\delta_{\mathrm{car}}, \delta_{\mathrm{cur}}, \delta_{\mathrm{hom}} > 0$*.*

6. *Exact-preserving cotangent realizations*

$$\widetilde{\Psi}^{\mathrm{root}}_{\mathrm{car}}, \qquad \widetilde{\Psi}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \widetilde{\Psi}^{\mathrm{root}}_{\mathrm{hom}}$$

into the cotangent bundles of the same configuration spaces $\mathcal{Q}_{\mathrm{car}}, \mathcal{Q}_{\mathrm{cur}}, \mathcal{Q}_{\mathrm{hom}}$*, producing the root Liouville one-forms*

$$\mathbf{\Lambda}^{\mathrm{root}}_{\mathrm{car}}, \qquad \mathbf{\Lambda}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \mathbf{\Lambda}^{\mathrm{root}}_{\mathrm{hom}}$$

and the closed root two-forms

$$\mathbf{\Xi}^{\mathrm{root}}_{\mathrm{car}}, \qquad \mathbf{\Xi}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \mathbf{\Xi}^{\mathrm{root}}_{\mathrm{hom}}.$$

Their pre-ambient restrictions are exactly the recorded pre-ambient Liouville one-forms and pre-ambient closed two-form geometry.

7. *Exact-preserving skew-Hermitian connection one-forms*

$$\mathcal{B}^{\mathrm{conn}}_{\mathrm{car}} \in \Omega^1(\mathcal{R}_{\mathrm{car}}; \mathfrak{u}(\mathscr{U}_{\mathrm{car}})), \qquad \mathcal{B}^{\mathrm{conn}}_{\mathrm{cur}} \in \Omega^1(\mathcal{R}_{\mathrm{cur}}; \mathfrak{u}(\mathscr{U}_{\mathrm{cur}})), \qquad \mathcal{B}^{\mathrm{conn}}_{\mathrm{hom}} \in \Omega^1(\mathcal{R}_{\mathrm{hom}}; \mathfrak{u}(\mathscr{U}_{\mathrm{hom}}))$$

and the induced exact-preserving unitary parallel transports

$$\mathbb{P}_{\mathrm{car}}, \qquad \mathbb{P}_{\mathrm{cur}}, \qquad \mathbb{P}_{\mathrm{hom}}$$

attached to piecewise C^1 root curves, identity on constant curves, multiplicative under concatenation, inverse under the corresponding root time reversals, and exact-preserving on the fixed branch. Their pre-ambient restrictions are exactly the recorded pre-ambient connection one-forms and induced parallel transports.

8. *Exact-preserving root shadow one-forms*

$$\varsigma^{\mathrm{root}}_{\mathrm{car}}, \qquad \varsigma^{\mathrm{root}}_{\mathrm{cur}}, \qquad \varsigma^{\mathrm{root}}_{\mathrm{hom}},$$

root rate one-forms

$$\boldsymbol{\lambda}^{\mathrm{root}}_{\mathrm{car}}, \qquad \boldsymbol{\lambda}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \boldsymbol{\lambda}^{\mathrm{root}}_{\mathrm{hom}},$$

root density and root flux source fields

$$\boldsymbol{q}^{\mathrm{root}}_{\mathrm{car}}, \qquad \boldsymbol{q}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \boldsymbol{q}^{\mathrm{root}}_{\mathrm{hom}}, \qquad \boldsymbol{j}^{\mathrm{root}}_{\mathrm{car}}, \qquad \boldsymbol{j}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \boldsymbol{j}^{\mathrm{root}}_{\mathrm{hom}},$$

and root boundary potentials

$$\boldsymbol{\rho}^{\partial,\mathrm{root}}_{\mathrm{car}}, \qquad \boldsymbol{\rho}^{\partial,\mathrm{root}}_{\mathrm{cur}}, \qquad \boldsymbol{\rho}^{\partial,\mathrm{root}}_{\mathrm{hom}}.$$

Their pre-ambient restrictions are exactly the recorded pre-ambient-basic shadow / rate / density / flux / boundary data.

9. *Fixed-endpoint root actions*

$$\mathscr{A}^{\mathrm{root}}_{\mathrm{car}}, \qquad \mathscr{A}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \mathscr{A}^{\mathrm{root}}_{\mathrm{hom}}$$

on curves constrained to the current pre-ambient exact-constraint shells inside the root bundles, whose stationary curves are exactly the curves singled out there by the restricted root Liouville / exact-constraint geometry, whose first variations vanish automatically along the characteristic, symmetry, vertical pre-ambient-fiber, and root-shell-normal directions, and whose second variations are positive on smooth endpoint-fixed complements to those directions and to the normal exact-constraint directions singled out by $\delta_{\mathrm{car}}, \delta_{\mathrm{cur}}, \delta_{\mathrm{hom}}$. These actions restrict to exactly the same fixed-endpoint pre-ambient rate-action geometry already recorded below them.

These data are exact fixed inputs for the present note. The only admissible positive result is to derive or justify why this full layer is itself forced while keeping it unchanged.

Definition 3 (Recorded exact-preservation target)**.** *The present note must preserve, without any*

modification,

$$\mathscr{F}_{\mathrm{sel}}, \qquad \mathscr{F}_{\mathrm{deep}}, \qquad \mathscr{F}_{\mathrm{car}}[\mathfrak{S},\mathfrak{R},\mathfrak{F}], \qquad \mathscr{F}_{\mathrm{prim}}[K,J,F], \qquad \mathscr{F}_{\mathrm{mic}}[H,Y], \tag{5}$$

together with the fixed free-sector verdict

$$\Pi_{\mathrm{free}}\widehat{\Sigma}^{\mathrm{cmp}} = 0, \tag{6}$$

and hence the same previously recorded $\Pi_{\mathrm{free}}H = 0$ *outcome, the same completion solve*

$$\Sigma^{\mathrm{cmp}} = \Sigma^{\mathrm{cmp}}, \tag{7}$$

the same one operative metric $g^{\sharp}$*, and the same recorded Phase D / E infrared-spectrum and universal-coupling verdicts together with the same Phase F controlled GR limit.*

3. Primitive Root-Generating Shell and Lifted Connection-Cotangent Origin of the Root Geometry

Definition 4 (Primitive root-generating shell / lifted connection-cotangent geometry)**.** *For each sharper sector, let*

$$\mathcal{P}_{\mathrm{car}}, \qquad \mathcal{P}_{\mathrm{cur}}, \qquad \mathcal{P}_{\mathrm{hom}}$$

be finite-dimensional C^2 *manifolds of yet more primitive exact-preserving substrate states. Assume:*

1. *there are smooth visible primitive readout maps*

$$\Pi^{\mathrm{prim}}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to \mathcal{O}_{\mathrm{car}}, \qquad \Pi^{\mathrm{prim}}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to \mathcal{O}_{\mathrm{cur}}, \qquad \Pi^{\mathrm{prim}}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to \mathcal{O}_{\mathrm{hom}},$$

   together with exact-preserving primitive unit sections

$$u^{\mathrm{prim}}_{\mathrm{car}} : \mathcal{O}_{\mathrm{car}} \to \mathcal{P}_{\mathrm{car}}, \qquad u^{\mathrm{prim}}_{\mathrm{cur}} : \mathcal{O}_{\mathrm{cur}} \to \mathcal{P}_{\mathrm{cur}}, \qquad u^{\mathrm{prim}}_{\mathrm{hom}} : \mathcal{O}_{\mathrm{hom}} \to \mathcal{P}_{\mathrm{hom}}$$

   satisfying

$$\Pi^{\mathrm{prim}}_{\mathrm{car}} \circ u^{\mathrm{prim}}_{\mathrm{car}} = \mathrm{id}_{\mathcal{O}_{\mathrm{car}}}, \qquad \Pi^{\mathrm{prim}}_{\mathrm{cur}} \circ u^{\mathrm{prim}}_{\mathrm{cur}} = \mathrm{id}_{\mathcal{O}_{\mathrm{cur}}}, \qquad \Pi^{\mathrm{prim}}_{\mathrm{hom}} \circ u^{\mathrm{prim}}_{\mathrm{hom}} = \mathrm{id}_{\mathcal{O}_{\mathrm{hom}}}.$$

2. *there are smooth involutive exact-preserving primitive time reversals*

$$\sigma^{\mathrm{prim}}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to \mathcal{P}_{\mathrm{car}}, \qquad \sigma^{\mathrm{prim}}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to \mathcal{P}_{\mathrm{cur}}, \qquad \sigma^{\mathrm{prim}}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to \mathcal{P}_{\mathrm{hom}}$$

   such that

$$\Pi^{\mathrm{prim}}_{\mathrm{car}} \circ \sigma^{\mathrm{prim}}_{\mathrm{car}} = \Pi^{\mathrm{prim}}_{\mathrm{car}}, \qquad \Pi^{\mathrm{prim}}_{\mathrm{cur}} \circ \sigma^{\mathrm{prim}}_{\mathrm{cur}} = \Pi^{\mathrm{prim}}_{\mathrm{cur}}, \qquad \Pi^{\mathrm{prim}}_{\mathrm{hom}} \circ \sigma^{\mathrm{prim}}_{\mathrm{hom}} = \Pi^{\mathrm{prim}}_{\mathrm{hom}},$$

   and

$$\sigma^{\mathrm{prim}}_{\mathrm{car}} \circ u^{\mathrm{prim}}_{\mathrm{car}} = u^{\mathrm{prim}}_{\mathrm{car}}, \qquad \sigma^{\mathrm{prim}}_{\mathrm{cur}} \circ u^{\mathrm{prim}}_{\mathrm{cur}} = u^{\mathrm{prim}}_{\mathrm{cur}}, \qquad \sigma^{\mathrm{prim}}_{\mathrm{hom}} \circ u^{\mathrm{prim}}_{\mathrm{hom}} = u^{\mathrm{prim}}_{\mathrm{hom}}.$$

3. *each primitive space carries a smooth exact-preserving positive-definite metric*

$$\breve{M}_{\mathrm{car}}, \qquad \breve{M}_{\mathrm{cur}}, \qquad \breve{M}_{\mathrm{hom}}$$

preserved by the corresponding primitive time reversals, and each visible primitive fiber

$$(\Pi^{\mathrm{prim}}_{\mathrm{car}})^{-1}(x), \qquad (\Pi^{\mathrm{prim}}_{\mathrm{cur}})^{-1}(y), \qquad (\Pi^{\mathrm{prim}}_{\mathrm{hom}})^{-1}(z)$$

is compact.

4. *there are finite-dimensional real root-generating target spaces*

$$\mathcal{W}^{\mathrm{prim}}_{\mathrm{car}}, \qquad \mathcal{W}^{\mathrm{prim}}_{\mathrm{cur}}, \qquad \mathcal{W}^{\mathrm{prim}}_{\mathrm{hom}}$$

with positive-definite metrics

$$q^{\mathrm{prim}}_{\mathrm{car}}, \qquad q^{\mathrm{prim}}_{\mathrm{cur}}, \qquad q^{\mathrm{prim}}_{\mathrm{hom}},$$

and exact-preserving smooth root-generating shell sections

$$\mathfrak{P}^{\mathrm{prim}}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to \mathcal{W}^{\mathrm{prim}}_{\mathrm{car}}, \qquad \mathfrak{P}^{\mathrm{prim}}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to \mathcal{W}^{\mathrm{prim}}_{\mathrm{cur}}, \qquad \mathfrak{P}^{\mathrm{prim}}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to \mathcal{W}^{\mathrm{prim}}_{\mathrm{hom}}$$

preserved by the corresponding primitive time reversals and vanishing on the corresponding primitive unit sections.

Define the recorded root substrate bundles as the exact zero shells

$$\mathcal{R}_{\mathrm{car}} \equiv \mathfrak{P}^{\mathrm{prim}^{-1}}_{\mathrm{car}}(0), \qquad \mathcal{R}_{\mathrm{cur}} \equiv \mathfrak{P}^{\mathrm{prim}^{-1}}_{\mathrm{cur}}(0), \qquad \mathcal{R}_{\mathrm{hom}} \equiv \mathfrak{P}^{\mathrm{prim}^{-1}}_{\mathrm{hom}}(0). \tag{8}$$

Let

$$\iota^{\mathrm{root}}_{\mathrm{car}} : \mathcal{R}_{\mathrm{car}} \hookrightarrow \mathcal{P}_{\mathrm{car}}, \qquad \iota^{\mathrm{root}}_{\mathrm{cur}} : \mathcal{R}_{\mathrm{cur}} \hookrightarrow \mathcal{P}_{\mathrm{cur}}, \qquad \iota^{\mathrm{root}}_{\mathrm{hom}} : \mathcal{R}_{\mathrm{hom}} \hookrightarrow \mathcal{P}_{\mathrm{hom}} \tag{9}$$

be the shell inclusions. Assume these shells are closed embedded C^2 submanifolds, invariant under the corresponding primitive time reversals, and containing the images of the corresponding primitive unit sections. Assume further that for every $r \in \mathcal{R}_{\mathrm{car}}$, $s \in \mathcal{R}_{\mathrm{cur}}$, $t \in \mathcal{R}_{\mathrm{hom}}$,

$$q^{\mathrm{prim}}_{\mathrm{car}}\Big(D\mathfrak{P}^{\mathrm{prim}}_{\mathrm{car}}(r)\xi, D\mathfrak{P}^{\mathrm{prim}}_{\mathrm{car}}(r)\xi\Big) \geq \lambda^{\mathrm{prim}}_{\mathrm{car}}\, \breve{M}_{\mathrm{car}}(\xi,\xi), \qquad \lambda^{\mathrm{prim}}_{\mathrm{car}} > 0, \tag{10}$$

$$q^{\mathrm{prim}}_{\mathrm{cur}}\Big(D\mathfrak{P}^{\mathrm{prim}}_{\mathrm{cur}}(s)\eta, D\mathfrak{P}^{\mathrm{prim}}_{\mathrm{cur}}(s)\eta\Big) \geq \lambda^{\mathrm{prim}}_{\mathrm{cur}}\, \breve{M}_{\mathrm{cur}}(\eta,\eta), \qquad \lambda^{\mathrm{prim}}_{\mathrm{cur}} > 0, \tag{11}$$

$$q^{\mathrm{prim}}_{\mathrm{hom}}\Big(D\mathfrak{P}^{\mathrm{prim}}_{\mathrm{hom}}(t)\zeta, D\mathfrak{P}^{\mathrm{prim}}_{\mathrm{hom}}(t)\zeta\Big) \geq \lambda^{\mathrm{prim}}_{\mathrm{hom}}\, \breve{M}_{\mathrm{hom}}(\zeta,\zeta), \qquad \lambda^{\mathrm{prim}}_{\mathrm{hom}} > 0, \tag{12}$$

for all vectors ξ, η, ζ orthogonal, with respect to the corresponding primitive metrics, to the tangent spaces of the corresponding zero shells.

5. *there are exact-preserving lifted pre-ambient-generating shell maps*

$$\widehat{\mathfrak{Z}}^{\mathrm{root}}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to \mathcal{Z}^{\mathrm{root}}_{\mathrm{car}}, \qquad \widehat{\mathfrak{Z}}^{\mathrm{root}}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to \mathcal{Z}^{\mathrm{root}}_{\mathrm{cur}}, \qquad \widehat{\mathfrak{Z}}^{\mathrm{root}}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to \mathcal{Z}^{\mathrm{root}}_{\mathrm{hom}}$$

preserved by the corresponding primitive time reversals and vanishing on the corresponding

primitive unit sections. Define the recorded pre-ambient-generating shell sections by restriction:

$$\mathfrak{Z}^{\mathrm{root}}_{\mathrm{car}} \equiv \widehat{\mathfrak{Z}}^{\mathrm{root}}_{\mathrm{car}}|_{\mathcal{R}_{\mathrm{car}}}, \qquad \mathfrak{Z}^{\mathrm{root}}_{\mathrm{cur}} \equiv \widehat{\mathfrak{Z}}^{\mathrm{root}}_{\mathrm{cur}}|_{\mathcal{R}_{\mathrm{cur}}}, \qquad \mathfrak{Z}^{\mathrm{root}}_{\mathrm{hom}} \equiv \widehat{\mathfrak{Z}}^{\mathrm{root}}_{\mathrm{hom}}|_{\mathcal{R}_{\mathrm{hom}}}. \tag{13}$$

Assume these restricted sections satisfy the same recorded positive root gap bounds $\kappa^{\mathrm{root}}_{\mathrm{car}}, \kappa^{\mathrm{root}}_{\mathrm{cur}}, \kappa^{\mathrm{root}}_{\mathrm{hom}} > 0$ *on the current pre-ambient shells cut out inside* $\mathcal{R}_{\mathrm{car}}, \mathcal{R}_{\mathrm{cur}}, \mathcal{R}_{\mathrm{hom}}$.

6. *there are exact-preserving lifted ambient-generating shell maps*

$$\widehat{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to \mathcal{H}^{\mathrm{orig}}_{\mathrm{car}}, \qquad \widehat{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to \mathcal{H}^{\mathrm{orig}}_{\mathrm{cur}}, \qquad \widehat{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to \mathcal{H}^{\mathrm{orig}}_{\mathrm{hom}}$$

preserved by the corresponding primitive time reversals and vanishing on the corresponding primitive unit sections. Define the recorded lifted ambient-generating shell maps by restriction:

$$\widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{car}} \equiv \widehat{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{car}}|_{\mathcal{R}_{\mathrm{car}}}, \qquad \widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{cur}} \equiv \widehat{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{cur}}|_{\mathcal{R}_{\mathrm{cur}}}, \qquad \widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{hom}} \equiv \widehat{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{hom}}|_{\mathcal{R}_{\mathrm{hom}}}. \tag{14}$$

Assume these restricted maps satisfy the same recorded positive ambient-generating gap bounds $\epsilon^{\mathrm{orig}}_{\mathrm{car}}, \epsilon^{\mathrm{orig}}_{\mathrm{cur}}, \epsilon^{\mathrm{orig}}_{\mathrm{hom}} > 0$ *on the current ambient shells cut out inside the current pre-ambient shells.*

7. *there are exact-preserving lifted shell maps*

$$\widehat{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to \mathcal{E}_{\mathrm{car}}, \qquad \widehat{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to \mathcal{E}_{\mathrm{cur}}, \qquad \widehat{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to \mathcal{E}_{\mathrm{hom}}$$

preserved by the corresponding primitive time reversals and vanishing on the corresponding primitive unit sections. Define the recorded lifted shell maps by restriction:

$$\widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{car}} \equiv \widehat{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{car}}|_{\mathcal{R}_{\mathrm{car}}}, \qquad \widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{cur}} \equiv \widehat{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{cur}}|_{\mathcal{R}_{\mathrm{cur}}}, \qquad \widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{hom}} \equiv \widehat{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{hom}}|_{\mathcal{R}_{\mathrm{hom}}}. \tag{15}$$

Assume these restricted maps satisfy the same recorded positive shell-gap bounds $\eta_{\mathrm{car}}, \eta_{\mathrm{cur}}, \eta_{\mathrm{hom}} > 0$ *on the current root exact-constraint shells.*

8. *there are exact-preserving lifted exact-constraint maps*

$$\widehat{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to \mathcal{K}_{\mathrm{car}}, \qquad \widehat{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to \mathcal{K}_{\mathrm{cur}}, \qquad \widehat{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to \mathcal{K}_{\mathrm{hom}}$$

preserved by the corresponding primitive time reversals and vanishing on the corresponding primitive unit sections. Define the recorded lifted exact-constraint maps by restriction:

$$\widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{car}} \equiv \widehat{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{car}}|_{\mathcal{R}_{\mathrm{car}}}, \qquad \widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{cur}} \equiv \widehat{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{cur}}|_{\mathcal{R}_{\mathrm{cur}}}, \qquad \widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{hom}} \equiv \widehat{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{hom}}|_{\mathcal{R}_{\mathrm{hom}}}. \tag{16}$$

9. *there are exact-preserving cotangent realizations*

$$\breve{\Psi}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to T^*\mathcal{Q}_{\mathrm{car}}, \qquad \breve{\Psi}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to T^*\mathcal{Q}_{\mathrm{cur}}, \qquad \breve{\Psi}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to T^*\mathcal{Q}_{\mathrm{hom}}$$

into the same recorded configuration spaces. Let θ_{can} *denote the canonical Liouville one-form on the relevant cotangent bundle. Define primitive Liouville one-forms and closed two-forms by*

$$\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{car}} \equiv (\breve{\Psi}_{\mathrm{car}})^*\theta_{\mathrm{can}}, \qquad \mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{cur}} \equiv (\breve{\Psi}_{\mathrm{cur}})^*\theta_{\mathrm{can}}, \qquad \mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{hom}} \equiv (\breve{\Psi}_{\mathrm{hom}})^*\theta_{\mathrm{can}}, \tag{17}$$

$$\mathbf{\Xi}^{\mathrm{prim}}_{\mathrm{car}} \equiv d\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{car}}, \qquad \mathbf{\Xi}^{\mathrm{prim}}_{\mathrm{cur}} \equiv d\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{cur}}, \qquad \mathbf{\Xi}^{\mathrm{prim}}_{\mathrm{hom}} \equiv d\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{hom}}. \tag{18}$$

Assume

$$(\sigma^{\mathrm{prim}}_{\mathrm{car}})^*\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{car}} = -\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{car}}, \qquad (\sigma^{\mathrm{prim}}_{\mathrm{cur}})^*\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{cur}} = -\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{cur}}, \qquad (\sigma^{\mathrm{prim}}_{\mathrm{hom}})^*\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{hom}} = -\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{hom}}.$$

Define the recorded root Liouville one-forms and closed two-forms by shell restriction:

$$\mathbf{\Lambda}^{\mathrm{root}}_{\mathrm{car}} \equiv (\iota^{\mathrm{root}}_{\mathrm{car}})^*\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{car}}, \qquad \mathbf{\Lambda}^{\mathrm{root}}_{\mathrm{cur}} \equiv (\iota^{\mathrm{root}}_{\mathrm{cur}})^*\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{cur}}, \qquad \mathbf{\Lambda}^{\mathrm{root}}_{\mathrm{hom}} \equiv (\iota^{\mathrm{root}}_{\mathrm{hom}})^*\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{hom}}, \tag{19}$$

$$\mathbf{\Xi}^{\mathrm{root}}_{\mathrm{car}} \equiv (\iota^{\mathrm{root}}_{\mathrm{car}})^*\mathbf{\Xi}^{\mathrm{prim}}_{\mathrm{car}}, \qquad \mathbf{\Xi}^{\mathrm{root}}_{\mathrm{cur}} \equiv (\iota^{\mathrm{root}}_{\mathrm{cur}})^*\mathbf{\Xi}^{\mathrm{prim}}_{\mathrm{cur}}, \qquad \mathbf{\Xi}^{\mathrm{root}}_{\mathrm{hom}} \equiv (\iota^{\mathrm{root}}_{\mathrm{hom}})^*\mathbf{\Xi}^{\mathrm{prim}}_{\mathrm{hom}}. \tag{20}$$

10. *there are smooth skew-Hermitian primitive connection one-forms*

$$\mathcal{C}^{\mathrm{conn}}_{\mathrm{car}} \in \Omega^1(\mathcal{P}_{\mathrm{car}}; \mathfrak{u}(\mathscr{U}_{\mathrm{car}})), \qquad \mathcal{C}^{\mathrm{conn}}_{\mathrm{cur}} \in \Omega^1(\mathcal{P}_{\mathrm{cur}}; \mathfrak{u}(\mathscr{U}_{\mathrm{cur}})), \qquad \mathcal{C}^{\mathrm{conn}}_{\mathrm{hom}} \in \Omega^1(\mathcal{P}_{\mathrm{hom}}; \mathfrak{u}(\mathscr{U}_{\mathrm{hom}}))$$

that are exact-preserving on the fixed branch and satisfy the time-reversal equivariance rules

$$(\sigma^{\mathrm{prim}}_{\mathrm{car}})^*\mathcal{C}^{\mathrm{conn}}_{\mathrm{car}} = -\mathcal{C}^{\mathrm{conn}}_{\mathrm{car}}, \qquad (\sigma^{\mathrm{prim}}_{\mathrm{cur}})^*\mathcal{C}^{\mathrm{conn}}_{\mathrm{cur}} = -\mathcal{C}^{\mathrm{conn}}_{\mathrm{cur}}, \qquad (\sigma^{\mathrm{prim}}_{\mathrm{hom}})^*\mathcal{C}^{\mathrm{conn}}_{\mathrm{hom}} = -\mathcal{C}^{\mathrm{conn}}_{\mathrm{hom}}.$$

Let

$$\mathbb{U}_{\mathrm{car}}(\Gamma) \in \mathcal{U}(\mathscr{U}_{\mathrm{car}}), \qquad \mathbb{U}_{\mathrm{cur}}(\Gamma) \in \mathcal{U}(\mathscr{U}_{\mathrm{cur}}), \qquad \mathbb{U}_{\mathrm{hom}}(\Gamma) \in \mathcal{U}(\mathscr{U}_{\mathrm{hom}})$$

denote the parallel transports generated by those primitive connection one-forms along piecewise C^1 primitive curves Γ. Define the recorded root connection one-forms and induced parallel transports by

$$\mathcal{B}^{\mathrm{conn}}_{\mathrm{car}} \equiv (\iota^{\mathrm{root}}_{\mathrm{car}})^*\mathcal{C}^{\mathrm{conn}}_{\mathrm{car}}, \qquad \mathcal{B}^{\mathrm{conn}}_{\mathrm{cur}} \equiv (\iota^{\mathrm{root}}_{\mathrm{cur}})^*\mathcal{C}^{\mathrm{conn}}_{\mathrm{cur}}, \qquad \mathcal{B}^{\mathrm{conn}}_{\mathrm{hom}} \equiv (\iota^{\mathrm{root}}_{\mathrm{hom}})^*\mathcal{C}^{\mathrm{conn}}_{\mathrm{hom}}, \tag{21}$$

$$\mathbb{P}_{\mathrm{car}}(\gamma) \equiv \mathbb{U}_{\mathrm{car}}(\iota^{\mathrm{root}}_{\mathrm{car}} \circ \gamma), \qquad \mathbb{P}_{\mathrm{cur}}(\gamma) \equiv \mathbb{U}_{\mathrm{cur}}(\iota^{\mathrm{root}}_{\mathrm{cur}} \circ \gamma), \qquad \mathbb{P}_{\mathrm{hom}}(\gamma) \equiv \mathbb{U}_{\mathrm{hom}}(\iota^{\mathrm{root}}_{\mathrm{hom}} \circ \gamma) \tag{22}$$

on piecewise C^1 root curves γ.

Define the recorded root readout maps, root unit sections, root time reversals, and root metrics by restriction:

$$\Pi^{\mathrm{root}}_{\mathrm{car}} \equiv \Pi^{\mathrm{prim}}_{\mathrm{car}} \circ \iota^{\mathrm{root}}_{\mathrm{car}}, \qquad \Pi^{\mathrm{root}}_{\mathrm{cur}} \equiv \Pi^{\mathrm{prim}}_{\mathrm{cur}} \circ \iota^{\mathrm{root}}_{\mathrm{cur}}, \qquad \Pi^{\mathrm{root}}_{\mathrm{hom}} \equiv \Pi^{\mathrm{prim}}_{\mathrm{hom}} \circ \iota^{\mathrm{root}}_{\mathrm{hom}}, \tag{23}$$

$$u^{\mathrm{root}}_{\mathrm{car}} \equiv u^{\mathrm{prim}}_{\mathrm{car}}, \qquad u^{\mathrm{root}}_{\mathrm{cur}} \equiv u^{\mathrm{prim}}_{\mathrm{cur}}, \qquad u^{\mathrm{root}}_{\mathrm{hom}} \equiv u^{\mathrm{prim}}_{\mathrm{hom}}, \tag{24}$$

$$\varrho^{\mathrm{root}}_{\mathrm{car}} \equiv \sigma^{\mathrm{prim}}_{\mathrm{car}}|_{\mathcal{R}_{\mathrm{car}}}, \qquad \varrho^{\mathrm{root}}_{\mathrm{cur}} \equiv \sigma^{\mathrm{prim}}_{\mathrm{cur}}|_{\mathcal{R}_{\mathrm{cur}}}, \qquad \varrho^{\mathrm{root}}_{\mathrm{hom}} \equiv \sigma^{\mathrm{prim}}_{\mathrm{hom}}|_{\mathcal{R}_{\mathrm{hom}}}, \tag{25}$$

$$\overline{M}_{\mathrm{car}} \equiv (\iota^{\mathrm{root}}_{\mathrm{car}})^*\breve{M}_{\mathrm{car}}, \qquad \overline{M}_{\mathrm{cur}} \equiv (\iota^{\mathrm{root}}_{\mathrm{cur}})^*\breve{M}_{\mathrm{cur}}, \qquad \overline{M}_{\mathrm{hom}} \equiv (\iota^{\mathrm{root}}_{\mathrm{hom}})^*\breve{M}_{\mathrm{hom}}. \tag{26}$$

For each visible point, define the visible root fibers by

$$(\Pi^{\mathrm{root}}_{\mathrm{car}})^{-1}(x) = \mathfrak{P}^{\mathrm{prim}\,-1}_{\mathrm{car}}(0) \cap (\Pi^{\mathrm{prim}}_{\mathrm{car}})^{-1}(x), \tag{27}$$

with the corresponding formulas in the cur *and* hom *sectors.*

Proposition 1 (Root substrate bundles, pre-ambient-generating shell sections, lifted ambient / shell / exact-constraint maps, cotangent geometry, and connection-transports are forced)**.** *The primitive root-generating shell / lifted connection-cotangent data of Definition 4 recover exactly the geometric part of the recorded root-shell / lifted connection-cotangent / parallel-transport layer.*

1. *The exact zero shells* (8) *are exactly the recorded root substrate bundles* $\mathcal{R}_{\mathrm{car}}, \mathcal{R}_{\mathrm{cur}}, \mathcal{R}_{\mathrm{hom}}$.

2. *The restricted maps* (23)–(26) *are exactly the recorded root readout maps, root unit sections, root time reversals, and preserved positive root metrics, and the visible root fibers* (27) *and their sector analogues are compact.*

3. *The restricted maps* (13) *are exactly the recorded pre-ambient-generating shell sections, and their root gap bounds are the same recorded bounds* $\kappa^{\mathrm{root}}_{\mathrm{car}}, \kappa^{\mathrm{root}}_{\mathrm{cur}}, \kappa^{\mathrm{root}}_{\mathrm{hom}}$ *on the current pre-ambient shells.*

4. *The restricted maps* (14) *are exactly the recorded lifted ambient-generating shell maps, and the recorded target metrics* $n^{\mathrm{orig}}_{\mathrm{car}}, n^{\mathrm{orig}}_{\mathrm{cur}}, n^{\mathrm{orig}}_{\mathrm{hom}}$ *are already the deeper target metrics on which those maps are defined.*

5. *The restricted maps* (15) *are exactly the recorded lifted shell maps, and the recorded shell target metrics* $m^{\mathrm{sh}}_{\mathrm{car}}, m^{\mathrm{sh}}_{\mathrm{cur}}, m^{\mathrm{sh}}_{\mathrm{hom}}$ *are already the deeper target metrics on which those maps are defined.*

6. *The restricted maps* (16) *are exactly the recorded lifted exact-constraint maps, and the recorded target metrics* $m^{\mathrm{con}}_{\mathrm{car}}, m^{\mathrm{con}}_{\mathrm{cur}}, m^{\mathrm{con}}_{\mathrm{hom}}$ *are already the deeper target metrics on which those maps are defined.*

7. *The restricted forms* (19)–(20) *are exactly the recorded root cotangent-model Liouville one-forms and closed two-form geometry.*

8. *The restricted connection one-forms and parallel-transport formulas* (21)–(22) *are exactly the recorded root connection one-forms and their induced parallel transports.*

9. *Consequently the root substrate bundles, their pre-ambient-generating shell sections, their lifted ambient-generating shell maps, their lifted shell maps and lifted exact-constraint maps with the same target metrics, their root cotangent-model Liouville one-forms and closed two-form geometry, and their root connection one-forms with induced parallel transports are no longer primitive at present scope; they are forced by a deeper primitive root-generating shell / lifted connection-cotangent layer.*

Proof. Item (1) is the root-generating shell construction itself. By assumption, the zero shells of the primitive shell sections are closed embedded C^2 submanifolds, invariant under the primitive time reversals, and containing the primitive units. Those are exactly the recorded root substrate bundles.

Item (2) is immediate from restriction. The primitive readouts already split the visible object classes, the primitive units are already sections of those readouts, the primitive time reversals already preserve the readouts and units, and the primitive metrics are positive and time-reversal invariant. Restricting to the exact zero shells gives exactly the recorded root objects. Each visible primitive fiber is compact by assumption, and each visible root fiber is its closed intersection with the root-generating shell, so the visible root fibers are compact as well.

Item (3) follows from the restriction formulas (13). No additional root-level pre-ambient-generating shell data are chosen. The lifted pre-ambient-generating shell maps already live on the deeper primitive spaces, and the recorded root shell sections are simply their restrictions to the root zero shells. The same root gap bounds on the current pre-ambient shells are kept fixed by assumption on those restricted maps.

Item (4) is the same restriction argument one step later. The recorded lifted ambient-generating shell maps are not inserted separately; they are already present as root-shell restrictions of deeper exact-preserving maps into the same target spaces equipped with the same target metrics.

Items (5) and (6) are the same restriction argument applied first to the lifted shell maps and then to the lifted exact-constraint maps. The recorded root maps are not inserted separately; they are already present as shell restrictions of deeper exact-preserving maps into the same target spaces equipped with the same target metrics.

Item (7) is the cotangent reduction statement. The primitive Liouville one-forms are pulled back from the canonical Liouville forms on the same recorded cotangent bundles, so they are not arbitrary inserted one-forms. Restricting them to the root zero shells gives exactly the recorded root Liouville one-forms, and restricting their exterior derivatives gives exactly the recorded root closed two-forms.

Item (8) follows from ordinary restriction and parallel transport. The primitive skew-Hermitian connection one-forms determine unitary transports on primitive curves, identity on constant curves, multiplicativity under concatenation, and inversion under time reversal. Restricting the one-forms to the root zero shells gives exactly the recorded root connection one-forms, and restricting the primitive transports to curves lying in those zero shells gives exactly the recorded root transports. Thus the current root connection data are not independent curve assignments; they are forced as shell restrictions.

Item (9) is therefore immediate: every geometric component of the recorded root-shell / lifted connection-cotangent / parallel-transport layer is recovered verbatim, but it is no longer primitive. □

4. Primitive-Basic Source and Fixed-Endpoint Origin of the Remaining Root Data

Definition 5 (Primitive-basic source and fixed-endpoint action data)**.** *Assume the primitive geometry of Definition 4 is fixed. Assume further:*

1. *there are exact-preserving primitive shadow one-forms*

$$\varsigma^{\mathrm{prim}}_{\mathrm{car}} \in \Omega^1(\mathcal{P}_{\mathrm{car}}; \mathcal{Y}^{\mathrm{st}}_{\mathrm{car}}), \qquad \varsigma^{\mathrm{prim}}_{\mathrm{cur}} \in \Omega^1(\mathcal{P}_{\mathrm{cur}}; \mathcal{Y}^{\mathrm{st}}_{\mathrm{cur}}), \qquad \varsigma^{\mathrm{prim}}_{\mathrm{hom}} \in \Omega^1(\mathcal{P}_{\mathrm{hom}}; \mathcal{Y}^{\mathrm{st}}_{\mathrm{hom}})$$

   that are invariant under the corresponding primitive time reversals, annihilate vectors normal to the root-generating shell loci $\mathcal{R}_{\mathrm{car}}, \mathcal{R}_{\mathrm{cur}}, \mathcal{R}_{\mathrm{hom}}$, annihilate vertical vectors tangent to the visible primitive fibers, and are preserved along the current root characteristic curves singled out on the root exact-constraint shells by the restricted root Liouville / exact-constraint geometry. Define the recorded root shadow one-forms by restriction:

$$\varsigma^{\mathrm{root}}_{\mathrm{car}} \equiv (\iota^{\mathrm{root}}_{\mathrm{car}})^* \varsigma^{\mathrm{prim}}_{\mathrm{car}}, \qquad \varsigma^{\mathrm{root}}_{\mathrm{cur}} \equiv (\iota^{\mathrm{root}}_{\mathrm{cur}})^* \varsigma^{\mathrm{prim}}_{\mathrm{cur}}, \qquad \varsigma^{\mathrm{root}}_{\mathrm{hom}} \equiv (\iota^{\mathrm{root}}_{\mathrm{hom}})^* \varsigma^{\mathrm{prim}}_{\mathrm{hom}}. \tag{28}$$

2. *there are exact-preserving primitive rate one-forms*

$$\boldsymbol{\lambda}^{\mathrm{prim}}_{\mathrm{car}} \in \Omega^1(\mathcal{P}_{\mathrm{car}}), \qquad \boldsymbol{\lambda}^{\mathrm{prim}}_{\mathrm{cur}} \in \Omega^1(\mathcal{P}_{\mathrm{cur}}), \qquad \boldsymbol{\lambda}^{\mathrm{prim}}_{\mathrm{hom}} \in \Omega^1(\mathcal{P}_{\mathrm{hom}})$$

   with the same invariance and annihilation properties. Define the recorded root rate one-forms by

$$\boldsymbol{\lambda}^{\mathrm{root}}_{\mathrm{car}} \equiv (\iota^{\mathrm{root}}_{\mathrm{car}})^* \boldsymbol{\lambda}^{\mathrm{prim}}_{\mathrm{car}}, \qquad \boldsymbol{\lambda}^{\mathrm{root}}_{\mathrm{cur}} \equiv (\iota^{\mathrm{root}}_{\mathrm{cur}})^* \boldsymbol{\lambda}^{\mathrm{prim}}_{\mathrm{cur}}, \qquad \boldsymbol{\lambda}^{\mathrm{root}}_{\mathrm{hom}} \equiv (\iota^{\mathrm{root}}_{\mathrm{hom}})^* \boldsymbol{\lambda}^{\mathrm{prim}}_{\mathrm{hom}}. \tag{29}$$

3. *there are exact-preserving primitive density and primitive flux source fields*

$$\boldsymbol{q}^{\mathrm{prim}}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to \mathscr{N}_{\mathrm{car}}, \qquad \boldsymbol{q}^{\mathrm{prim}}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to \mathscr{N}_{\mathrm{cur}}, \qquad \boldsymbol{q}^{\mathrm{prim}}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to \mathscr{N}_{\mathrm{hom}},$$

$$\boldsymbol{j}^{\mathrm{prim}}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to \mathscr{J}_{\mathrm{car}}, \qquad \boldsymbol{j}^{\mathrm{prim}}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to \mathscr{J}_{\mathrm{cur}}, \qquad \boldsymbol{j}^{\mathrm{prim}}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to \mathscr{J}_{\mathrm{hom}},$$

invariant under the corresponding primitive time reversals, preserved along the current root characteristic curves, and constant on root-shell-normal directions and on vertical primitive-fiber directions. Define the recorded root density and root flux source fields by restriction:

$$\boldsymbol{q}^{\mathrm{root}}_{\mathrm{car}} \equiv \boldsymbol{q}^{\mathrm{prim}}_{\mathrm{car}}|_{\mathcal{R}_{\mathrm{car}}}, \qquad \boldsymbol{q}^{\mathrm{root}}_{\mathrm{cur}} \equiv \boldsymbol{q}^{\mathrm{prim}}_{\mathrm{cur}}|_{\mathcal{R}_{\mathrm{cur}}}, \qquad \boldsymbol{q}^{\mathrm{root}}_{\mathrm{hom}} \equiv \boldsymbol{q}^{\mathrm{prim}}_{\mathrm{hom}}|_{\mathcal{R}_{\mathrm{hom}}}, \tag{30}$$

$$\boldsymbol{j}^{\mathrm{root}}_{\mathrm{car}} \equiv \boldsymbol{j}^{\mathrm{prim}}_{\mathrm{car}}|_{\mathcal{R}_{\mathrm{car}}}, \qquad \boldsymbol{j}^{\mathrm{root}}_{\mathrm{cur}} \equiv \boldsymbol{j}^{\mathrm{prim}}_{\mathrm{cur}}|_{\mathcal{R}_{\mathrm{cur}}}, \qquad \boldsymbol{j}^{\mathrm{root}}_{\mathrm{hom}} \equiv \boldsymbol{j}^{\mathrm{prim}}_{\mathrm{hom}}|_{\mathcal{R}_{\mathrm{hom}}}. \tag{31}$$

4. *there are exact-preserving primitive boundary potentials*

$$\boldsymbol{\rho}^{\partial,\mathrm{prim}}_{\mathrm{car}} : \mathcal{P}_{\mathrm{car}} \to \mathscr{R}_{\mathrm{car}}, \qquad \boldsymbol{\rho}^{\partial,\mathrm{prim}}_{\mathrm{cur}} : \mathcal{P}_{\mathrm{cur}} \to \mathscr{R}_{\mathrm{cur}}, \qquad \boldsymbol{\rho}^{\partial,\mathrm{prim}}_{\mathrm{hom}} : \mathcal{P}_{\mathrm{hom}} \to \mathscr{R}_{\mathrm{hom}}$$

vanishing on the corresponding primitive unit sections, invariant under the corresponding primitive time reversals, annihilating vectors normal to the root-generating shells, and constant on vertical primitive-fiber directions. Define the recorded root boundary potentials by

$$\boldsymbol{\rho}^{\partial,\mathrm{root}}_{\mathrm{car}} \equiv \boldsymbol{\rho}^{\partial,\mathrm{prim}}_{\mathrm{car}}|_{\mathcal{R}_{\mathrm{car}}}, \qquad \boldsymbol{\rho}^{\partial,\mathrm{root}}_{\mathrm{cur}} \equiv \boldsymbol{\rho}^{\partial,\mathrm{prim}}_{\mathrm{cur}}|_{\mathcal{R}_{\mathrm{cur}}}, \qquad \boldsymbol{\rho}^{\partial,\mathrm{root}}_{\mathrm{hom}} \equiv \boldsymbol{\rho}^{\partial,\mathrm{prim}}_{\mathrm{hom}}|_{\mathcal{R}_{\mathrm{hom}}}. \tag{32}$$

5. *define the primitive fixed-endpoint rate actions on C^2 curves constrained to the current root exact-constraint shells inside the root-generating shells by*

$$\mathscr{A}^{\mathrm{prim}}_{\mathrm{car}}[\Gamma] \equiv \int_{\Gamma} \boldsymbol{\lambda}^{\mathrm{prim}}_{\mathrm{car}}, \tag{33}$$

$$\mathscr{A}^{\mathrm{prim}}_{\mathrm{cur}}[\Gamma] \equiv \int_{\Gamma} \boldsymbol{\lambda}^{\mathrm{prim}}_{\mathrm{cur}}, \tag{34}$$

$$\mathscr{A}^{\mathrm{prim}}_{\mathrm{hom}}[\Gamma] \equiv \int_{\Gamma} \boldsymbol{\lambda}^{\mathrm{prim}}_{\mathrm{hom}}. \tag{35}$$

Assume their stationary curves at fixed visible endpoint data are exactly the current root characteristic curves determined on those root exact-constraint shells by the restricted root Liouville / exact-constraint geometry, that their first variations vanish automatically along the characteristic, symmetry, vertical root-fiber, and root-shell-normal directions, and that their second variations are positive on smooth endpoint-fixed complements to those directions and to the normal exact-constraint directions singled out by $\delta_{\mathrm{car}}, \delta_{\mathrm{cur}}, \delta_{\mathrm{hom}}$. Define the recorded root fixed-endpoint actions by

$$\mathscr{A}^{\mathrm{root}}_{\mathrm{car}}[\gamma] \equiv \int_{\gamma} \boldsymbol{\lambda}^{\mathrm{root}}_{\mathrm{car}}, \tag{36}$$

$$\mathscr{A}^{\mathrm{root}}_{\mathrm{cur}}[\gamma] \equiv \int_{\gamma} \boldsymbol{\lambda}^{\mathrm{root}}_{\mathrm{cur}}, \tag{37}$$

$$\mathscr{A}^{\mathrm{root}}_{\mathrm{hom}}[\gamma] \equiv \int_{\gamma} \boldsymbol{\lambda}^{\mathrm{root}}_{\mathrm{hom}}. \tag{38}$$

Proposition 2 (Root-basic data and root action are forced)**.** *The primitive basic data of Definition 5 recover exactly the remaining recorded root-shell / lifted connection-cotangent / parallel-transport input layer.*

1. *The restricted one-forms* (28) *and* (29) *are exactly the recorded root shadow and root rate one-forms.*

2. *The restricted source fields* (30) *and* (31) *are exactly the recorded root density and root flux source fields.*

3. *The restricted boundary potentials* (32) *are exactly the recorded root boundary potentials.*

4. *The fixed-endpoint actions* (36)–(38) *are exactly the recorded root actions, and their inherited second-variation positivity is exactly the recorded fixed-endpoint root rate-action geometry.*

5. *Consequently the root-basic shadow / rate / density / flux / boundary data and the fixed-endpoint root rate-action geometry are no longer primitive at present scope; they are forced by deeper primitive-basic restriction and the same fixed-endpoint primitive variational geometry.*

Proof. Item (1) follows from the restriction construction itself. Because the primitive shadow and primitive rate one-forms annihilate both the root-shell-normal directions and the vertical directions tangent to the visible primitive fibers, their restrictions to the root zero shells are well defined and already satisfy the invariance and horizontality required at root level.

Item (2) is immediate from the restriction formulas (30) and (31). Constancy on root-shell-normal directions and on vertical primitive-fiber directions guarantees that the restricted source fields depend only on the root states themselves, while the assumed characteristic-flow and time-reversal invariance give exactly the recorded exact-preserving behavior.

Item (3) is the same argument applied to the boundary potentials. Since the primitive boundary potentials vanish on the primitive unit sections, their root restrictions vanish on the recorded root unit sections as required.

Item (4) follows from fixed-endpoint stationary reduction. The primitive rate actions are assumed to have precisely the current root characteristic curves as stationary curves, with first variations vanishing automatically along the characteristic, symmetry, vertical root-fiber, and root-shell-normal directions. Restricting those actions to curves in the root zero shells yields exactly the recorded root-action formulas (36)–(38). The positive second variation on endpoint-fixed complements descends directly to the recorded root stationary geometry.

Item (5) is therefore immediate. None of the current root-basic data or root fixed-endpoint stationary geometry needs to be inserted separately once the primitive-basic layer is fixed. □

5. Exact Recovery of the Recorded Root Layer and Downstream Package

Theorem 1 (Same root-shell / lifted connection-cotangent / parallel-transport input layer)**.** *Under the fixed branch of Definition 1, the primitive root-generating shell / lifted connection-cotangent construction recovers exactly the root-shell / lifted connection-cotangent / parallel-transport input layer of Definition 2.*

Proof. The geometric part is Proposition 1. The remaining basic-data and fixed-endpoint-action part is Proposition 2. Together they recover every item of Definition 2 without alteration. □

Theorem 2 (Same pre-ambient layer and same downstream package)**.** *Under the fixed branch of Definition 1, the deeper primitive root-generating shell / lifted connection-cotangent construction preserves without change:*

1. *the same pre-ambient-shell / connection-cotangent / parallel-transport input layer;*
2. *the same ambient-shell / cotangent-Liouville / connection-holonomy input layer;*
3. *the same constrained substrate-bundle / exact-constraint / Liouville-holonomy input layer;*
4. *the same proto-process / stationary-selection / characteristic-bundle input layer;*
5. *the same process-state / reversible-line-field / compact-calibration-fiber / basic-form layer;*
6. *the same reversible-process-thread / calibration-generator / shadow-current / rate-density / boundary-recorder layer;*
7. *the same reversible-process-window / calibration-action / stationary-shadow / local-rate / endpoint-recorder layer;*
8. *the same reversible-history / stationary-equivalence / short-history-action / endpoint-readout / cotangent-lift layer;*
9. *the same reversible-groupoid / transport-action / constrained-stationary package;*
10. *the same reconfiguration-group / transport-law / equilibrium-reduction package;*
11. *the same derivation-algebra / balance-law / equilibrium-manifold layer;*
12. *hence the same reversible process-thread / process-window / history / groupoid / reconfiguration / derivation chain;*
13. *the same* $\mathscr{F}_{\mathrm{sel}}$*;*
14. *the same exact-preservation normal form* $\mathscr{F}_{\mathrm{deep}}$*;*
15. *the same carrier-origin functional* $\mathscr{F}_{\mathrm{car}}[\mathfrak{S}, \mathfrak{R}, \mathfrak{F}]$*;*
16. *the same primitive reservoir* $\mathscr{F}_{\mathrm{prim}}[K, J, F]$*;*
17. *the same microscopic-equilibrium reservoir* $\mathscr{F}_{\mathrm{mic}}[H, Y]$*;*
18. *the same free-sector verdict* $\Pi_{\mathrm{free}}\widehat{\Sigma}^{\mathrm{cmp}} = 0$*, and hence the same* $\Pi_{\mathrm{free}}H = 0$ *outcome;*
19. *the same substrate and observer completion solves* (4)*;*
20. *the same* Σ^{cmp} *solve, the same observer-equation closure, the same one operative metric* $g^{\sharp}$*, and the same recorded Phase D / E and Phase F gains.*

Proof. Theorem 1 reproduces exactly the input layer required by the immediately preceding pre-ambient-shell / connection-cotangent / parallel-transport origin note. That note already proves that once this root input layer is fixed, the same pre-ambient layer follows without change, and with it the same full downstream chain. Since none of the visible slots, elimination steps, or completion equations are altered here, every downstream object listed above is reproduced verbatim. □

Corollary 1 (Recorded gain package is preserved)**.** *Because the same downstream package is recovered, the recorded Phase D / E infrared-spectrum and universal-coupling verdicts, the recorded Phase F controlled GR limit, and the already recorded observer-equation closure package remain preserved without modification.*

Proof. This is the direct content of Theorem 2. □

6. Claim-Boundary Verdict

Theorem 3 (Root-shell / lifted connection-cotangent / parallel-transport origin verdict)**.** *At the disciplined scope of the present note, the exact positive result is the following.*

1. *The exact-preserving root substrate bundles* $\mathcal{R}_{\mathrm{car}}, \mathcal{R}_{\mathrm{cur}}, \mathcal{R}_{\mathrm{hom}}$ *are no longer primitive at current scope.*

2. *They arise as the exact zero shells of deeper root-generating sections* $\mathfrak{P}^{\mathrm{prim}}_{\mathrm{car}}, \mathfrak{P}^{\mathrm{prim}}_{\mathrm{cur}}, \mathfrak{P}^{\mathrm{prim}}_{\mathrm{hom}}$ *on compact visible primitive substrate bundles.*

3. *The visible root readout maps* $\Pi^{\mathrm{root}}_{\mathrm{car}}, \Pi^{\mathrm{root}}_{\mathrm{cur}}, \Pi^{\mathrm{root}}_{\mathrm{hom}}$*, the root unit sections* $u^{\mathrm{root}}_{\mathrm{car}}, u^{\mathrm{root}}_{\mathrm{cur}}, u^{\mathrm{root}}_{\mathrm{hom}}$*, the root time reversals* $\varrho^{\mathrm{root}}_{\mathrm{car}}, \varrho^{\mathrm{root}}_{\mathrm{cur}}, \varrho^{\mathrm{root}}_{\mathrm{hom}}$*, and the preserved positive root metrics* $\overline{M}_{\mathrm{car}}, \overline{M}_{\mathrm{cur}}, \overline{M}_{\mathrm{hom}}$ *are no longer primitive either.*

4. *They are the zero-shell restrictions of deeper primitive readouts, units, time reversals, and primitive metrics, and the recorded compact visible root fibers are closed shell slices inside compact visible primitive fibers.*

5. *The pre-ambient-generating shell sections* $\mathfrak{Z}^{\mathrm{root}}_{\mathrm{car}}, \mathfrak{Z}^{\mathrm{root}}_{\mathrm{cur}}, \mathfrak{Z}^{\mathrm{root}}_{\mathrm{hom}}$ *are no longer primitive either.*

6. *They arise as the root-shell restrictions of deeper lifted pre-ambient-generating shell maps* $\widehat{\mathfrak{Z}}^{\mathrm{root}}_{\mathrm{car}}, \widehat{\mathfrak{Z}}^{\mathrm{root}}_{\mathrm{cur}}, \widehat{\mathfrak{Z}}^{\mathrm{root}}_{\mathrm{hom}}$ *into the same positive target spaces with the same target metrics* $p^{\mathrm{root}}_{\mathrm{car}}, p^{\mathrm{root}}_{\mathrm{cur}}, p^{\mathrm{root}}_{\mathrm{hom}}$.

7. *The lifted ambient-generating shell maps* $\widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{car}}, \widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{cur}}, \widetilde{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{hom}}$ *are no longer primitive either.*

8. *They arise as the root-shell restrictions of deeper lifted ambient-generating shell maps* $\widehat{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{car}}, \widehat{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{cur}}, \widehat{\mathfrak{S}}^{\mathrm{orig}}_{\mathrm{hom}}$ *into the same positive target spaces with the same target metrics* $n^{\mathrm{orig}}_{\mathrm{car}}, n^{\mathrm{orig}}_{\mathrm{cur}}, n^{\mathrm{orig}}_{\mathrm{hom}}$.

9. *The lifted shell maps* $\widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{car}}, \widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{cur}}, \widetilde{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{hom}}$ *are no longer primitive either.*

10. *They arise as the root-shell restrictions of deeper lifted shell maps* $\widehat{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{car}}, \widehat{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{cur}}, \widehat{\mathfrak{D}}^{\mathrm{root}}_{\mathrm{hom}}$ *into the same positive target spaces with the same target metrics* $m^{\mathrm{sh}}_{\mathrm{car}}, m^{\mathrm{sh}}_{\mathrm{cur}}, m^{\mathrm{sh}}_{\mathrm{hom}}$.

11. *The lifted exact-constraint maps* $\widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{car}}, \widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{cur}}, \widetilde{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{hom}}$ *and their target metrics* $m^{\mathrm{con}}_{\mathrm{car}}, m^{\mathrm{con}}_{\mathrm{cur}}, m^{\mathrm{con}}_{\mathrm{hom}}$ *are no longer primitive either.*

12. *They arise as the root-shell restrictions of deeper lifted exact-constraint maps* $\widehat{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{car}}, \widehat{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{cur}}, \widehat{\mathfrak{C}}^{\mathrm{root}}_{\mathrm{hom}}$ *into the same positive target spaces with the same target metrics.*

13. *The root cotangent-model Liouville one-forms* $\mathbf{\Lambda}^{\mathrm{root}}_{\mathrm{car}}, \mathbf{\Lambda}^{\mathrm{root}}_{\mathrm{cur}}, \mathbf{\Lambda}^{\mathrm{root}}_{\mathrm{hom}}$ *and the root closed two-form geometry* $\mathbf{\Xi}^{\mathrm{root}}_{\mathrm{car}}, \mathbf{\Xi}^{\mathrm{root}}_{\mathrm{cur}}, \mathbf{\Xi}^{\mathrm{root}}_{\mathrm{hom}}$ *are no longer primitive either.*

14. *They arise as the root-shell restrictions of deeper cotangent-model Liouville one-forms* $\mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{car}}, \mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{cur}}, \mathbf{\Lambda}^{\mathrm{prim}}_{\mathrm{hom}}$ *and their exact two-forms on the primitive bundles.*

15. *The root connection one-forms* $\mathcal{B}^{\mathrm{conn}}_{\mathrm{car}}, \mathcal{B}^{\mathrm{conn}}_{\mathrm{cur}}, \mathcal{B}^{\mathrm{conn}}_{\mathrm{hom}}$ *and their induced parallel transports* $\mathbb{P}_{\mathrm{car}}, \mathbb{P}_{\mathrm{cur}}, \mathbb{P}_{\mathrm{hom}}$ *are no longer primitive either.*

16. *They arise as root-shell restrictions of deeper skew-Hermitian primitive connection one-forms* $\mathcal{C}^{\mathrm{conn}}_{\mathrm{car}}, \mathcal{C}^{\mathrm{conn}}_{\mathrm{cur}}, \mathcal{C}^{\mathrm{conn}}_{\mathrm{hom}}$ *and of their induced primitive parallel transports.*

17. *The root-basic shadow / rate / density / flux / boundary data and the fixed-endpoint root rate-action geometry are no longer primitive either.*

18. *They arise as root-shell restrictions of deeper primitive-basic source data and the same fixed-endpoint primitive rate-action geometry.*

19. *Therefore the full root-shell / lifted connection-cotangent / parallel-transport input layer of the immediately preceding note is recovered without change.*

20. *Consequently the same pre-ambient-shell / connection-cotangent / parallel-transport layer, the same ambient-shell / cotangent-Liouville / connection-holonomy layer, the same process-state layer, the same reversible process-thread / process-window / history / groupoid / reconfiguration / derivation chain, the same* $\mathscr{F}_{\mathrm{sel}}$*, the same exact-preservation normal form, the same carrier-origin / primitive-reservoir / microscopic-equilibrium functionals, the same* Σ^{cmp} *solve, the same* $\Pi_{\mathrm{free}} H = 0$ *outcome, the same observer-equation closure, the same one operative metric, and the same recorded Phase D / E and Phase F gains remain unchanged.*

21. *Stronger promotion language is still not justified here, because the present note explains only why the previous root-shell / lifted connection-cotangent / parallel-transport layer arises from a deeper primitive root-generating shell / lifted connection-cotangent layer; it does not yet derive why that new deeper primitive layer is itself unique, final, or inevitable beyond current scope.*

22. *No broader packaging consequence or default side-line continuation follows from the mathematical content of this note alone. The remaining honest burden is one layer deeper again on this same exact-preservation line, or else another comparably concrete observer-equation gain on the same one-metric route.*

Proof. Items (1)–(16) are Proposition 1. Items (17) and (18) are Proposition 2. Items (19) and (20) are Theorems 1 and 2 together with Corollary 1. Items (21) and (22) are the present claim-boundary discipline stated in the abstract and introduction. □

Corollary 2 (What remains next)**.** *The primary active burden moves one layer deeper again. It is no longer to justify why the root substrate bundles, their pre-ambient-generating shell sections, their lifted ambient-generating shell maps, their lifted shell maps and lifted exact-constraint maps with the same target metrics, their root cotangent-model Liouville one-forms and closed two-form geometry, their connection one-forms and induced parallel transports, or their root-basic shadow / rate / density / flux / boundary data are admissible at current scope; that step is now recorded. The next honest burden is either to derive or justify why the deeper primitive substrate bundles, their root-generating shell sections, their lifted pre-ambient-generating shell maps, their lifted ambient-generating shell maps, their lifted shell and exact-constraint maps, their cotangent-model Liouville one-forms, their connection one-forms and parallel transports, and their primitive-basic source / action data are themselves forced by yet more primitive substrate structure, or else to produce another comparably*

concrete observer-equation gain on the same one-metric line. No stronger promotion language, no packaging pass, and no default move onto the anchored positive-pair side line are justified unless they directly discharge one of those burdens.

7. Conclusion

The positive result of this note is that the root-shell / lifted connection-cotangent / parallel-transport layer is no longer left primitive at present scope. It is recovered from a still deeper primitive root-generating shell / lifted connection-cotangent layer.

At that deeper level, the recorded root substrate bundles

$$\mathcal{R}_{\mathrm{car}}, \qquad \mathcal{R}_{\mathrm{cur}}, \qquad \mathcal{R}_{\mathrm{hom}}$$

are exact zero shells inside compact visible primitive substrate bundles. Their readout maps, unit sections, time reversals, and preserved positive metrics are zero-shell restrictions of deeper primitive data. Their pre-ambient-generating shell sections are root-shell restrictions of deeper lifted pre-ambient-generating shell maps into the same positive target spaces with the same target metrics. Their lifted ambient-generating shell maps are root-shell restrictions of deeper lifted ambient-generating shell maps into the same positive target spaces with the same target metrics. Their lifted shell maps and lifted exact-constraint maps are root-shell restrictions of deeper lifted maps into the same target spaces with the same target metrics. Their root cotangent-model Liouville one-forms and closed two-form geometry are root-shell restrictions of deeper cotangent-model Liouville data. Their root connection one-forms and induced parallel transports are root-shell restrictions of deeper skew-Hermitian primitive connection one-forms and their primitive parallel transports. Their root-basic shadow / rate / density / flux / boundary data and their fixed-endpoint root rate-action geometry are root-shell restrictions of deeper primitive-basic data and the same primitive fixed-endpoint variational geometry.

Because that deeper construction reproduces the full root-shell / lifted connection-cotangent / parallel-transport input layer without changing any of its visible slots, the immediately preceding pre-ambient-shell / connection-cotangent / parallel-transport note applies verbatim. Consequently the same pre-ambient-shell / connection-cotangent / parallel-transport layer, the same ambient-shell / cotangent-Liouville / connection-holonomy layer, the same process-state layer, the same reversible process-thread / process-window / history / groupoid / reconfiguration / derivation chain, the same $\mathscr{F}_{\mathrm{sel}}$, the same exact-preservation normal form, the same carrier-origin / primitive-reservoir / microscopic-equilibrium functionals, the same Σ^{cmp} solve, the same $\Pi_{\mathrm{free}} H = 0$ outcome, the same observer-equation closure, the same one operative metric, and the same recorded Phase D / E and Phase F gains remain unchanged.

The scope is still disciplined rather than final. The present note does not prove that the deeper primitive root-generating shell / lifted connection-cotangent layer is unique or ultimate. The next honest burden is to derive or justify why that new deeper primitive layer itself is forced by still more primitive substrate structure, or else to record another comparably concrete observer-equation gain on the same one-metric line.